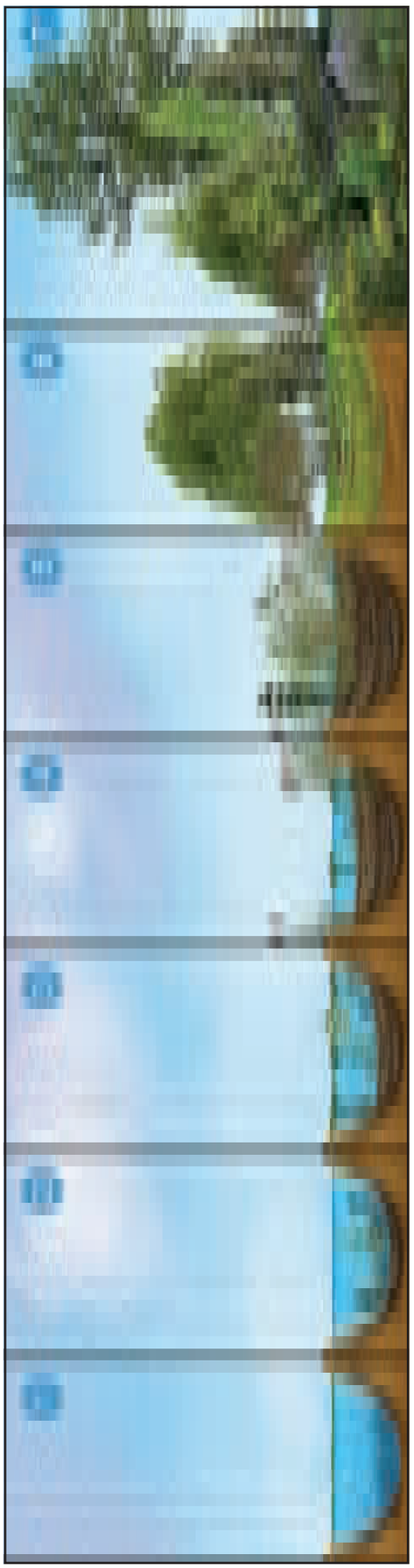


Figure 7.21: Diagrammatic representation shows different stages of hydrosere.

Carex, Juncus, Cyperus and *Eleocharis* colonise the area. They form a mat-like vegetation with the help of their much branched root system. This leads to an absorption and loss of large quantity of water. At the end of this stage, the soil becomes dry and the marshy vegetation disappears gradually and leads to shurb stage.

6. Shrub stage - As the disappearance of marshy vegetation continues, soil becomes dry. Hence, these areas are now invaded by terrestrial plants like shrubs (*Salix* and *Cornus*) and trees (*Populus* and *Alnus*). These plants absorb large quantity of water and make the habitat dry. Further, the accumulation of humus with a rich flora of microorganisms produce minerals in the soil, ultimately favouring the arrival of new tree species in the area.

7. Forest stage - It is the climax community of hydrosere. A variety of trees invade the area and develop any one of the diverse type of vegetation. Example:Temperate mixed forest (*Ulmus,Acer* and *Quercus*), Tropical rain forest (*Artocarpus* and *Cinnamomum*) and Tropical deciduous forest (*Bamboo* and *Tectona*).

In the 7 stages of hydrosere succession, stage1 is occupied by pioneer community, while the stage 7 is occupied by the climax community. The stages 2 to 6 are occupied by seral communities.

Lithosere

Lithosere is a type of xerosere initiating on a barren rock surface. The barren rock is devoid of water and organic matter. A barren rock surface gets mineral deposits due to weathering. This results in the colonization of pioneer organisms like crustose lichens. Through a series of successive seral stages, forest stage (Climax community) is achieved finally. These series of stages are given below Fig 7.22.

1. Crustose lichen stage - The pioneers like crustose lichens (*Rhizocarpon* and *Lecanora*) secrete some acids which enhance the

TN_GOVT_BOTANY_XII_PAGES_162-190 CH 07.indd 166 3/8/2020 2:54:05 PM